

Nodular and/or Conglobate Acne

Separate treatment suggestions are provided based on gender (see Fig. 4).

- In **males**, first-line therapy includes **oral isotretinoin** or a **topical fixed combination plus an oral antibiotic**.
- In **females**, either of these regimens may be supplemented with **oral anti-androgenic hormonal therapy**.^[55,60]
- **Females of childbearing age** prescribed oral isotretinoin must use **effective contraception** due to its **teratogenic potential**.^[61]
- An **initial high dose of antibiotic** should be considered in all patients.^[62]
- A minimum of **12–16 weeks** of treatment is typically required to achieve significant improvement.
- Treatment **dose and duration** should be individualized based on response.
- Once a good response is achieved, **maintenance therapy** with a **topical retinoid/BPO combination** can be considered.^[63]

Principal Reasons for Lack of Treatment Response

The main causes of treatment failure and corresponding management strategies are summarized in Tables 2–4.

Disease Progression

A lack of response in patients with **predominant comedonal, mild, or moderate papulopustular acne** may be due to **disease progression** to a more severe form, necessitating a change in therapeutic approach (refer to Tables 2, 3a).

Non-Drug-Related Reasons

Common non-drug-related factors contributing to treatment failure include:

- **Severe seborrhoea:** May reduce the contact time of topical agents on the skin. Management options include **oral isotretinoin** (in males and females) or **oral anti-androgenic hormonal therapy** (in females).[61,64]
- **Use of comedogenic products:** Greasy skincare or makeup products can exacerbate acne. Patients should use **non-comedogenic, oil-free moisturizers and cosmetics**.
- **Exposure to acne-inducing agents:** Such as dioxins, anabolic steroids, or certain medications (e.g., anti-epileptics). These should be avoided when possible.[1]
- **Lifestyle and dietary factors:** Stress, smoking, high glycaemic index diets, and dairy consumption have been associated with worsening acne. [1,2,65]
- **Acne cosmetica:** Should be considered in patients with **predominant comedonal acne** linked to cosmetic product use.

Additional Considerations

- Evaluate patient adherence and proper application techniques.
- Assess for **contact dermatitis** or **photosensitivity**.
- Avoid overuse of cleansers; recommend a **gentle cleanser with a pH around 5**, free of **α - or β -hydroxy acids**.
- Review all skincare products for potential irritants or comedogenic ingredients.

Note: Oral hormonal therapies (e.g., oral contraceptives containing cyproterone acetate, drospirenone, chlormadinone acetate) and androgen receptor antagonists (e.g., spironolactone) are options for females, particularly when oral contraceptives are contraindicated.[51–56] These therapies primarily target **excess sebum production** and **follicular keratinization**, and should be

combined with **topical fixed combination therapy** to address all four pathogenic factors of acne.[53,55]

Maintenance therapy recommendations align with those for **predominant comedonal acne**.